

PRACTICAL WEEK -1

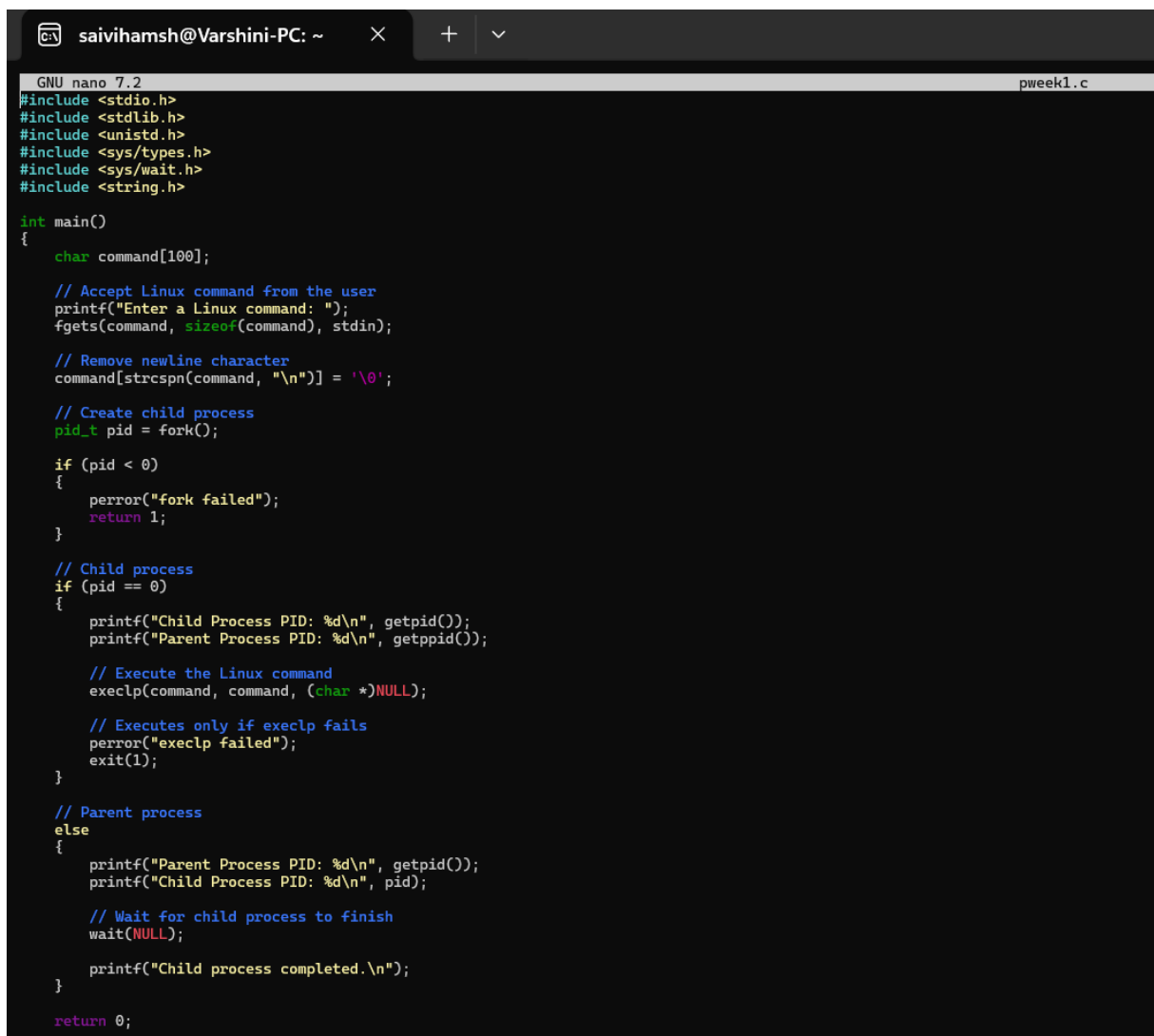
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Session-1

Task1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that:

1. Accept a Linux command as input.
2. Create a child process using fork().
3. Execute the command in the child process using an appropriate exec() system call.
4. Allow the parent process to wait for the child using wait().
5. Display the Process ID (PID) of both parent and child processes.



```
GNU nano 7.2 pweek1.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char command[100];

    // Accept Linux command from the user
    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove newline character
    command[strcspn(command, "\n")] = '\0';

    // Create child process
    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    // Child process
    if (pid == 0)
    {
        printf("Child Process PID: %d\n", getpid());
        printf("Parent Process PID: %d\n", getppid());

        // Execute the Linux command
        execlp(command, command, (char *)NULL);

        // Executes only if execlp fails
        perror("execlp failed");
        exit(1);
    }

    // Parent process
    else
    {
        printf("Parent Process PID: %d\n", getpid());
        printf("Child Process PID: %d\n", pid);

        // Wait for child process to finish
        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```

saiviamsh@Varshini-PC:~$ nano pweek1.c
saiviamsh@Varshini-PC:~$ gcc pweek1.c -o pweek1
saiviamsh@Varshini-PC:~$ ./pweek1
Enter a Linux command: ls
Parent Process PID: 502
Child Process PID: 505
Child Process PID: 505
Parent Process PID: 502
execvp failed: No such file or directory
Child process completed.

```

Observation: The program successfully accepted a Linux command as input and created a child process using `fork()`. The child process executed the entered `ls` command using `exec()`, while the parent process waited for the child process using `wait()`. The Parent PID and Child PID were displayed successfully, and the child process completed execution successfully.

Task2: Using Linux terminal commands (`uname`, `lscpu`, `lsblk`, `ps`, `top`), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```

saiviamsh@Varshini-PC:~$ pwd
/home/saiviamsh
saiviamsh@Varshini-PC:~$ uname -a
Linux Varshini-PC 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
saiviamsh@Varshini-PC:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              GenuineIntel
Model name:              13th Gen Intel(R) Core(TM) i7-1355U
CPU family:              6
Model:                  186
Thread(s) per core:      2
Core(s) per socket:      6
Socket(s):               1
Stepping:                3
BogoMIPS:                5222.39
Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht
                        syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_kno
                        wn_freq mmi1 pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes x
                        save avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow
                        ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx snap clflushopt clwb sha
                        _ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir6
                        4b fsrm md_clear serialize ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization:          VT-x
Hypervisor vendor:        Microsoft
Virtualization type:      full
Caches (sum of all):
  L1d:                     288 KiB (6 instances)
  L1i:                     192 KiB (6 instances)
  L2:                       7.5 MiB (6 instances)
  L3:                      12 MiB (1 instance)
NUMA:
NUMA node(s):             1
NUMA node0 CPU(s):        0-11
Vulnerabilities:
Gather data sampling:      Not affected
Ghostwrite:                Not affected
Indirect target selection: Not affected
Itlb multihit:             Not affected
L1tf:                      Not affected
Mds:                       Not affected
Meltdown:                  Not affected
Mmio stale data:           Not affected
Old microcode:             Not affected
Reg file data sampling:    Mitigation; Clear Register File
Retbleed:                  Mitigation; Enhanced IBRS
Spec rstack overflow:      Not affected
Spec store bypass:         Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:                Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:                Mitigation; Enhanced / Automatic IBRS; IBPB conditional; IBRSB=+IBRS SW sequence; BHI BHI_DIS_S
Srbds:                     Not affected
Taa:                       Not affected
Tsx async abort:           Not affected
Vmscape:                   Not affected
saiviamsh@Varshini-PC:~$

```

```
saiviamsh@Varshini-PC:~$ lsblk
NAME
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/

saiviamsh@Varshini-PC:~$ ps
PID TTY          TIME CMD
615 pts/0        00:00:00 bash
638 pts/0        00:00:00 ps
saiviamsh@Varshini-PC:~$
```

```
638 pts/0    00:00:00 ps
saiviamsh@Varshini-PC:~$ top
top - 20:49:56 up 28 min, 1 user, load average: 0.01, 0.04, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7796.6 total,  7231.6 free,   519.0 used,  196.2 buff/cache
MiB Swap: 2048.0 total,  2048.0 free,    0.0 used.  7277.6 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   21532 13048  9716 S   0.0   0.2   0:01.34 systemd
    2 root        20   0    3180  2208  2072 S   0.0   0.0   0:00.08 init-systemd(Ub
    6 root        20   0    3196  2128  2008 S   0.0   0.0   0:00.00 init
   58 root        19  -1  66820 17580 16468 S   0.0   0.2   0:00.41 systemd-journal
  104 root        20   0   25284  6644  5144 S   0.0   0.1   0:00.42 systemd-udev
  114 systemd+    20   0   21344 13184 11060 S   0.0   0.2   0:00.11 systemd-resolve
  115 systemd+    20   0   91036  7992  7024 S   0.0   0.1   0:00.17 systemd-timesyn
  178 root        20   0    4244  2680  2432 S   0.0   0.0   0:00.02 cron
  179 message+    20   0    9596  5388  4708 S   0.0   0.1   0:00.10 dbus-daemon
  193 root        20   0   17976  8880  7844 S   0.0   0.1   0:00.16 systemd-logind
  195 root        20   0 1756104 13376 11012 S   0.0   0.2   0:00.26 wsl-pro-service
  203 syslog      20   0  222516  5980  4576 S   0.0   0.1   0:00.15 rsyslogd
  219 root        20   0    3124  1976  1836 S   0.0   0.0   0:00.01 agetty
  229 root        20   0 107032 23344 13888 S   0.0   0.3   0:00.18 unattended-upgr
  314 root        20   0    6700  4680  3896 S   0.0   0.1   0:00.01 login
  366 saiviha+    20   0   20112 11380  9448 S   0.0   0.1   0:00.10 systemd
  367 saiviha+    20   0   21168  3576  1848 S   0.0   0.0   0:00.00 (sd-pam)
  395 saiviha+    20   0    6080  5320  3656 S   0.0   0.1   0:00.02 bash
  607 root        20   0    3184  1120   980 S   0.0   0.0   0:00.00 SessionLeader
  608 root        20   0    3200  1256  1108 S   0.0   0.0   0:00.00 Relay(615)
  615 saiviha+    20   0    6080  5420  3700 S   0.0   0.1   0:00.03 bash
  641 saiviha+    20   0    9300  5692  3512 R   0.0   0.1   0:00.00 top
```

```
saivihamsh@Varshini-PC:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.6Gi       2.4Mi       5.2Gi       3.5Mi       196Mi       7.1Gi
Swap:          2.0Gi          0B        2.0Gi

saivihamsh@Varshini-PC:~$ ./task2.sh

=====
TASK 2 - OS Hardware Investigation
=====

1. Operating System Information
-----
Linux Varshini-PC 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux

2. CPU Information
-----
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i7-1355U
CPU family:             6
Model:                 186
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               3
BogoMIPS:               5222.39
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
cp ln constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdpseed adx smap clflush
opt clwb sha_ni xsaveopt xsavec xgetbv1 xsave avx_vnni vmmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
clear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full

Caches (sum of all):
L1d:                    288 KiB (6 instances)
L1i:                    192 KiB (6 instances)
L2:                     7.5 MiB (6 instances)
L3:                     12 MiB (1 instance)

=====
3. Running Processes (First 10)
=====
UID          PID    PPID  C STIME TTY          TIME CMD
root          1        0 0 20:21 ?        00:00:01 /sbin/init
root          2        1 0 20:21 hvnc0 00:00:00 /init
root          6        2 0 20:21 hvnc0 00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root         58        1 0 20:21 ?        00:00:00 /usr/lib/systemd/systemd-journald
root         104        1 0 20:21 ?        00:00:00 /usr/lib/systemd/systemd-udevd
systemd+     114        1 0 20:21 ?        00:00:00 /usr/lib/systemd/systemd-resolved
systemd+     115        1 0 20:21 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
root         178        1 0 20:21 ?        00:00:00 /usr/sbin/cron -f -P
message+     179        1 0 20:21 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only

4. Real-Time System Monitoring (5 Seconds)
-----
top - 20:54:04 up 32 min, 1 user, load average: 0.03, 0.03, 0.00
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.3 us, 0.0 sy, 0.0 ni, 98.4 id, 0.8 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7796.6 total, 7228.3 free, 522.0 used, 196.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7274.7 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        0   0 21532 13048  9716 S   0.0   0.2   0:01.35 systemd
    2 root        0   0 3180 2208  2072 S   0.0   0.0   0:00.08 init-systemd(Ub
    6 root        0   0 3196 2128  2008 S   0.0   0.0   0:00.00 init
   58 root       19  -1 66820 17584 16472 S   0.0   0.2   0:00.43 systemd-journal
  104 root       20   0 25284 6604  5140 S   0.0   0.1   0:00.40 systemd-udevd
  114 systemd+   20   0 21344 13184 11060 S   0.0   0.2   0:00.12 systemd-resolve
  115 systemd+   20   0 91036 7992  7024 S   0.0   0.1   0:00.17 systemd-timesyn
  178 root       20   0 4244 2680  2432 S   0.0   0.0   0:00.03 cron
  179 message+    20   0 9156 5388  4708 S   0.0   0.1   0:00.11 dbus-daemon
  193 root       20   0 17976 8880  7244 S   0.0   0.1   0:00.16 systemd-logind
  195 root       20   0 1756104 13416 11012 S   0.0   0.2   0:00.28 wsl-pro-service
  203 syslog    20   0 222516 5980  4576 S   0.0   0.1   0:00.15 rsyslogd
  219 root       20   0 3124 1976  1836 S   0.0   0.0   0:00.01agetty
  229 root       20   0 107032 23344 13888 S   0.0   0.3   0:00.18 unattended-upgr
  314 root       20   0 6700 4680  3896 S   0.0   0.1   0:00.01 login
  366 saivih+    20   0 20112 11380  9448 S   0.0   0.1   0:00.10 systemd
  367 saivih+    20   0 21168 3576  1848 S   0.0   0.0   0:00.00 (sd-pam)
  395 saivih+    20   0 6080 5320  3656 S   0.0   0.1   0:00.02 bash
  607 root       20   0 3184 1120  980 S   0.0   0.0   0:00.00 SessionLeader
  608 root       20   0 3200 1256  1108 S   0.0   0.0   0:00.01 Relay(615)
  615 saivih+    20   0 6080 5420  3700 S   0.0   0.1   0:00.03 bash
  654 saivih+    20   0 4760 3404  3216 S   0.0   0.0   0:00.00 task2.sh
  659 saivih+    20   0 9292 5332  3240 R   0.0   0.1   0:00.00 top

=====
End of Report
=====
saivihamsh@Varshini-PC:~$ |
```

Observation: The Linux terminal commands successfully provided information about the system's hardware resources and operating system services. The uname command displayed kernel and system information, while lscpu showed CPU architecture, 12 CPUs, 6 cores, and 2 threads per core. The lsblk command displayed the available storage devices and swap space. The ps and top commands showed running processes and their resource usage. The OS abstracts CPU, memory, storage, and I/O devices by managing them through system services and providing applications with controlled interfaces instead of requiring direct access to the hardware.